OM RAJ

Agricultural Researcher • Environmental Analyst • Software Developer

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PROFESSIONAL SUMMARY

Interdisciplinary researcher, environmental analyst, and software developer with an **M.Sc. in Botany (Plant Pathology)** and hands-on project research experience at **IIT Ropar**. Uniquely blends wet-lab experimental rigor (spectrophotometry, chromatography, plant biochemical stress markers, wastewater testing) with full-stack desktop software engineering and digital agriculture knowledge systems. Proven record in executing government-sponsored research projects (PPCB and Punjab Excise Department), authoring peer-reviewed literature, developing production-ready scientific software deployed on the Microsoft Store, and designing digital public platforms. Equipped to drive impact across agri-tech knowledge engineering, environmental compliance monitoring, and computational research pipelines.

RESEARCH, TECHNICAL & LABORATORY SKILLS

Analytical & Spectroscopy Instruments:
ICP-MS (Inductively Coupled Plasma Mass Spectrometry), UV-Visible Spectrophotometer, Gas Chromatography (GC), Ion Chromatography, TVOC Analyzer, Carbon Analyzer, Aerosol Spectrophotometer, Formaldehyde Analyzer.

Regulatory Auditing & Environmental Compliance:
Environmental monitoring, industrial compliance audits (CPCC, PPCB, Excise Dept.), wastewater testing (CETP, STP, ETP), hazardous waste leachate classification, and compliance reporting.

Software Development & Computational Tech:
JavaScript, HTML5, CSS3, Electron.js, Node.js basics, REST API integration, Git/GitHub, Visual Studio Code.

Soil & Solid Waste Characterization:
Toxicity Characteristic Leaching Procedure (TCLP), acid digestion protocols, solid waste leachate extraction, heavy metal mobility profiling, bio-soil quality assessment, soil organic matter and nutrient testing.

Agriculture & Plant Pathology:
Crop disease diagnostics, pathogen identification, IDM, plant stress physiology, antioxidant enzyme assays (SOD, CAT, POD, proline, MDA), biomass/biochar soil amendments, soil fertility profiling.

Data Science, GIS & Creative Design:
MS Excel (formulas, linear regression), Power BI dashboards, QGIS, Adobe Photoshop, Canva Pro, Premiere Pro, CapCut, graphical abstracts for scientific journals, SOP authoring.

PROFESSIONAL RESEARCH & WORK EXPERIENCE

Project Researcher – Environmental Monitoring & Soil Audit

Dept. of Civil Engineering, IIT Ropar | Sponsor: Chandigarh Pollution Control Committee (CPCC) | PI: Dr. Indramani Dhada

- **Field Sampling & Soil Verification:** Field reconnaissance and composite sampling of legacy bio-soil from Dadumajra Dumping Site across Chandigarh urban reclamation sites (Sector 47 & 50).
- **Heavy Metal Trace Analysis (ICP-MS):** Solid matrix acid digestion and trace elemental quantification via Inductively Coupled Plasma Mass Spectrometry (ICP-MS).
- **Leachability & Toxicity (TCLP):** Toxicity Characteristic Leaching Procedure (TCLP) extractions evaluating heavy metal mobility and CPCB regulatory compliance.

- **Soil Quality & Audit Report:** Evaluated pH, EC, organic content, and reactivity to assess environmental hazards and compile the compliance audit report.

Project Researcher – Sustainable Agriculture & Environmental Monitoring
2025 – Present

Dept. of Civil Engineering, IIT Ropar | Supervisors: Dr. Indramani Dhada, Dr. Neelkanth Nirmalkar | PM: Dr. Harsh Sharma

- **Sustainable Agriculture & Biochar:** Investigated comparative impact of pineapple peel, jackfruit peel, and drek wood biochars on soil physicochemical characteristics, nutrient retention, and crop physiological responses.
- **Plant Biochemical Assays:** Quantified defense enzyme kinetics (SOD, CAT, POD, Proline, MDA) to evaluate plant resilience under abiotic and heavy-metal stress.
- **Wastewater & Industrial Audits (PPCB & Excise Dept.):** Conducted wet-chemical testing across CETPs, STPs, and Sutlej River basin; participated in industrial audits of distilleries evaluating Zero Liquid Discharge (ZLD) performance.

Postgraduate Dissertation Researcher — River Water Quality & Runoff Assessment
2024 – 2025

Patna University | Supervisor: Prof. Dr. Meenakshi Singh (Head, Dept. of Biotechnology)

- Conducted systematic multi-point field sampling along major ghats of Ganga River at Patna; analyzed physicochemical/biological parameters against WHO, CPCB, and BIS 10500 standards.

Medical Writing Intern & Freelance Media Specialist
2021 – Present

YuvaIntern (NSDC / Henry Harvin, Nov 2025) & Freelance / MARKAD LIMITED (2021–Present)

- Synthesized biomedical/environmental research reviews with standardized citations (APA, IEEE, Vancouver). Designed graphical abstracts, academic branding, and university election outreach media.

EDUCATION & ACADEMIC QUALIFICATIONS

Degree / Program	Specialization / Discipline	Institution / University	Year	Score / CGPA
M.Sc. in Botany	Plant Pathology	Department of Botany, Patna University	2025	**7.56 CGPA**
B.Sc. in Botany	Plant Sciences & Chemistry	Patna Science College, Patna University	2023	63.50%
Diploma in Computer Applications (DCA)	Software & Computing Systems	Vinita Computer Education	2022	64.40%
Senior Secondary (Class XII, Science)	Physics, Chemistry, Biology (PCB)	Shanti Niketan Sr. Sec. School (CBSE)	2019	58.40%
Secondary School (Class X)	General Science & Mathematics	Sunshine Prep. / High School (CBSE)	2017	81.70%

SOFTWARE ENGINEERING & PRODUCTION INNOVATIONS

- **AWE LAB Academic Portal (IIT Ropar):** Architected, designed, and deployed official laboratory web portal for Dr. Indramani Dhada's team at IIT Ropar. (AWE LAB Portal)
- **SNIFF DOI — Academic Literature Discovery Engine (Microsoft Store):** Desktop software indexing 250M+ articles. Features 6-tier research gap synthesis and CSL engine (APA, IEEE, Harvard, Vancouver). (Microsoft Store)
- **WASH (Water Analysis Suite Hub) — Microsoft Store:** Environmental compliance engine automating APHA 24th Ed & BIS 10500 algorithms, spectrophotometer linear regression calibration, and NABL PDF reports. (Microsoft Store)
- **Calculator Luxe & Clock Span (Watch Span):** High-precision scientific desktop calculator & lab reaction kinetics timer published on Microsoft Store.

PUBLICATIONS & RESEARCH MANUSCRIPTS

- **Peer-Reviewed Journal Article:** Raj, O., & Ambreen, D. (2026). Assessment of Physicochemical Parameters of Water Samples from the Ganga River at Patna. *International Journal of Versatile Research and Analysis (IJVRA)*, 3(4), 784–810. ISSN: 2984-8903. Paper ID: IJVRA2603368.
- **Research Preprint (Air Quality):** Raj, O., & Dhada, I. Spatiotemporal Assessment of Ambient Air Quality in Selected Urban, Tourist and Industrial Regions of Himachal Pradesh, India: A Case Study. *Research Square*. DOI: 10.21203/rs.3.rs-10492149/v1
- **Manuscript in Preparation (Sustainable Agriculture):** Raj, O., & Dhada, I. Comparative Assessment of Pineapple Peel, Jackfruit Peel, and Drek Wood Biochar on Plant Growth, Soil Fertility, and Biochemical Responses. (Evaluates SOD, CAT, POD, proline, MDA kinetics).

WORKSHOPS & PROFESSIONAL CERTIFICATIONS

- **Fundamentals of Remote Sensing:** NASA "Train - Empower - Advance" Initiative (Sept 2026) – Satellite observations & spatial datasets.
- **Advances in Genome Biology & Gene Regulation:** School of Life Sciences, JNU, New Delhi (June 2025).
- **Marine Data 4 Asia 2025:** Copernicus Marine Service & Mercator Ocean International, EU Program (June 2025).
- **Power BI & AI Interactive Dashboards:** OfficeMaster (Sept 2025).
- **Cancer Genomics, Clinical Research & AI:** Biotecnika (Sept 2025).

LEADERSHIP & EXTENSION ACTIVITIES

- **Event & Sports Coordinator:** Patna Science College (2023, 2024, 2025). Managed logistics, budgeting, and operations for 3 consecutive Annual Sports Meets. Recipient of *The Catalyst Award (2025)*.
- **Volunteer Coordinator:** Patna University Alumni Meet (2023, 2025). Managed guest reception, VIP protocol, and stage management.

PROFESSIONAL REFERENCES

Dr. Indramani Dhada
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